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A ballistocardiogram dataset with reference ECG signals for bed-based heart rhythm assessment

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The unobtrusive collection of ballistocardiogram (BCG) data makes it a promising option for continuous, low-effort cardiovascular monitoring. Key heart rhythm metrics such as heart rate (HR), inter-beat interval (IBI), and heart rate variability (HRV) are widely recognized indicators of cardiovascular health. The primary clinical goal of BCG applications is to accurately estimate these metrics. Most previous studies have primarily examined healthy young adults, which contrasts significantly with high-risk populations, such as older adults or cardiac patients, who would benefit the most from long-term monitoring in everyday settings. To bridge this gap, we collected nighttime BCG recordings from 46 middle-aged and elderly participants during natural sleep. The subjects included individuals with atrial arrhythmias, ventricular arrhythmias, myocardial ischemia, and nonclinical sinus arrhythmias. This dataset also contains synchronized Holter monitor recordings, which may enhance the use of BCG in assessing cardiac rhythms and detecting early cardiovascular diseases, while also testing the generalizability of BCG algorithms in real-world scenarios involving various health conditions.

Background & Summary

Continuous cardiovascular health monitoring enables early identification of risk factors and reduces the incidence of cardiovascular diseases and their complications[1]. Although the electrocardiogram (ECG) is a commonly used diagnostic tool, its reliance on skin contact for measurements limits its practical applications. First, long-term or daily monitoring with an ECG is inconvenient. Second, ECG may cause discomfort or skin irritation for some sensitive individuals[2]. Consequently, unobtrusive cardiovascular monitoring methods have drawn extensive interest in the research community. Among these methods, the ballistocardiogram (BCG) is a promising alternative to the ECG. BCG measures mechanical vibrations caused by the heart's contraction action and resulting blood flow, capturing subtle body movements linked to cardiovascular activities. These signals systematically reflect circulatory function, providing a basis for evaluating cardiovascular health[3–5].

Unlike ECG, which requires direct skin contact via electrodes, BCG devices can operate unobtrusively through everyday article surfaces[5,6] such as mattresses or chair cushions, eliminating the need for skin contact. Previous studies demonstrate BCG's versatility in applications ranging from early cardiovascular disease screening[7–10] and sleep monitoring/staging[11,12] to estimating ECG-derived metrics[13,14], biometric identification[15,16], emotion recognition[17], detecting respiratory disorders[18], and tracking behavioral patterns[19].

Heart rate (HR), inter-beat interval (IBI), and heart rate variability (HRV) are critical metrics for assessing cardiovascular risk[20,21] and form the basis of BCG-based cardiac rhythm analysis[22–36] (see Table 1). While developments in BCG technology have enabled monitoring of these indicators, current research primarily focuses on healthy young populations. These young and healthy subjects sharply contrast with clinical realities, where middle-aged and elderly individuals (50 years and older) represent the primary demographic needing continuous cardiovascular monitoring. Epidemiological studies reveal that the prevalence of cardiovascular diseases in this older group is 3 to 5 times higher than in younger adults[37,38]. Conditions such as atrial fibrillation and premature ventricular contractions can alter ventricular ejection patterns and induce morphological changes

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Metrics	References	Size	Length	Age (years)	Collection Environment	Cardiovascular Health Status
HR	Mora *et al.*[22]	16	5 min/p (avg.)	22–65 (avg. 30.2)	Controlled lab	No known CVD
		14	5 min/p (avg.)	22–46 (avg. 28.1)	Controlled lab	Not specified
	Liu *et al.*[23]	25	5 min/p (avg.)	30.5 ± 5.2	Controlled lab	Not specified
	Jung *et al.*[24]	11	5 min/p (avg.)	27.8 ± 4.0	Controlled lab	Not specified
		14	≥3 min/p	26.2 ± 3.0	Controlled lab	No known CVD
	Jiao *et al.*[25]	40	≥13 min/p	29.2 ± 6.2	Controlled lab	Not specified
	M. Zhang *et al.*[26]	78	7–14 h/p	22–79	Controlled lab	Not specified
IBI	Shen *et al.*[27]	14	≥5 min/p	24–34	Controlled lab	No known CVD
	Mai *et al.*[28]	43	30 min/p (avg.)	30.0 ± 10.2	Controlled lab	No known CVD 6 cases with suspected tachycardia
	Thirion *et al.*[29]	12	18 min/p (avg.)	22–56	Controlled lab	No known CVD
	García-Limón *et al.*[30]	7	80 s/p (avg.)	33 ± 6	Controlled lab	Not specified
		14	50 s/p (avg.)	34 ± 11	Controlled lab	Not specified
HRV	Zhao *et al.*[31]	34	5 min/p (avg.)	21–56	Not specified	No known CVD
	Gao & Zhao[32]	10	10 min/p (avg.)	18–22	Not specified	1 case of hypertension
	Lyu *et al.*[33]	5	5 min/p (avg.)	25–54	Controlled lab	1 case of premature beat
	C. Wu *et al.*[34]	24	20 min/p (avg.)	27 ± 6	Controlled lab	No known CVD
	B. Yu *et al.*[35]	37	8.5 h/p (avg.)	avg. 47.6	Free Clinical	Not specified, but all exhibited varying severity of SAS
	He *et al.*[36]	15	1 h/p (avg.)	Over 18	Controlled lab	No known CVD

Table 1. Overview of datasets utilized in BCG-related research on HR, IBI, and HRV.

References	Size	Length	Age (years)	Collection Environment	Cardiovascular Health Status	ECG
Carlson *et al.*[41]	40	5min/p(avg.)	18-65 (avg. 33.9)	Controlled lab	No known CVD	✓
Li *et al.*[42]	32	19–27 h/p	19–27	Free dormitory	No known CVD	×
Cimr *et al.*[43]	16	18–19 h/p	Not specified	Geracomium & ICU	Impending death	×
Studnicka *et al.*(a)[44]	20	8–9 h/p	Not specified	Controlled lab	Not specified	×
Ladrova *et al.*[47]	27	10–15 min/p(avg.)	21–29 (avg. 23.63)	Not specified	No known CVD	✓

Table 2. Overview of public BCG datasets.

in BCG waveforms, such as amplitude attenuation and interval irregularities[39,40]. These changes undermine the generalizability of existing BCG-based cardiac rhythm models, typically trained on homogeneous datasets from healthy subjects. Two systemic challenges exacerbate this situation:

- **Dataset Limitations**: Most BCG studies rely on proprietary datasets with narrow demographic representation, such as young adults without comorbidities. Public benchmarks that include pathological variations are lacking. For instance, current datasets rarely incorporate nocturnal BCG recordings from elderly patients with myocardial ischemia or arrhythmias, despite their clinical relevance.
- **Algorithmic Sensitivity**: Traditional BCG analysis pipelines, designed for stable sinus rhythms, struggle to distinguish noise from genuine pathological waveform changes in older populations. For example, irregular RR intervals caused by atrial fibrillation can distort HRV metrics such as the Standard Deviation of NN intervals (SDNN) and Root Mean Square of Successive Differences (RMSSD), which assume quasi-stationary signals.

We present the first open-access BCG dataset that captures nocturnal cardiac activity under real-world conditions. Our data were collected from 46 participants with varying cardiovascular health statuses, including arrhythmias, myocardial ischemia, and non-pathological variations, using unobtrusive piezoelectric film sensors placed under bedsheets. To enable realistic nighttime monitoring, participants were allowed to maintain their natural habits, including routine activities such as mobile device use and conversations, as well as sleep-related movements like turning over and bathroom visits. Importantly, all BCG signals were paired with synchronized three-lead Holter ECG recordings, providing a clinically validated, time-aligned reference for cardiac electrical activity. Compared to the existing public BCG datasets[41–47] (Table 2), our work offers three key features:

Clinically Diverse Cohort: The dataset includes participants across a spectrum of cardiovascular health, from healthy individuals with occasional sinus arrhythmias to patients diagnosed with arrhythmias (e.g., atrial fibrillation) and myocardial ischemia (Table 3). Each recording is paired with detailed diagnostic reports and timestamped annotations of cardiovascular events, enabling pathology-specific analysis.

Real-World Validity: Data were collected during natural overnight sleep in home-like settings, capturing behaviors such as movement, device use, and bathroom visits. This mirrors real-world algorithm deployment environments, ensuring robust testing of generalizability across unpredictable scenarios.

Major Categories of CVD	Subcategories of CVD
Atrial Arrhythmias	Atrial Fibrillation, Atrial Flutter, Atrial Tachycardia, Premature Atrial Contractions
Ventricular Arrhythmias	Premature Ventricular Contractions
Sinus Arrhythmias	Sinus Tachycardia, Sinus Bradycardia
Myocardial Ischemia	ST-Segment Depression

Table 3. Types of cardiovascular diseases Included in the dataset.

Synchronized ECG Reference: Simultaneous Holter ECG recordings provide precise, time-aligned cardiac electrical activity data.

This dataset supports the development and validation of BCG-based tools for:

- Refining heart rhythm metrics extraction (e.g., IBI, HRV), with emphasis on robustness to arrhythmias and waveform anomalies.
- Advancing early detection of cardiovascular disorders, including atrial fibrillation, atrial flutter, and ischemic events.
- Enabling multimodal fusion-based applications such as ECG synthesis, biometric identification, and longitudinal vital sign monitoring.

Methods

Participants. Our dataset comprises 46 participants (26 male, 20 female) aged 27–93 years. To ensure the subjects' demographic and clinical distributions are relevant to the monitoring objectives, we prioritized the individuals with diagnosed cardiovascular disease (CVD), including arrhythmias and ischemic heart disease, alongside healthy controls, reflecting the age and health diversity of populations requiring long-term cardiovascular monitoring. Before enrollment, all participants underwent structured medical history reviews to confirm diagnoses and stratify groups proportionally. This approach balances pathological and non-pathological cases while aligning with real-world clinical populations.

Participants were not excluded based on non-cardiovascular comorbidities (e.g., diabetes, hyperlipidemia, or Parkinson's disease). All individuals met predefined functional criteria, including preserved cognitive function and the ability to perform independent nighttime activities such as using electronic devices, turning over, or getting out of bed. These standards aim to replicate real-world user behavior in bed, ensuring data authenticity and applicability to practical monitoring scenarios.

All participants provided written informed consent after receiving a detailed explanation of the study's purpose, procedures, and data-sharing protocols. The experimental protocol, which included the public dissemination of anonymized data, received approval from the Ethics Committee of Run Run Shaw Hospital at Zhejiang University School of Medicine (Approval No. 20190520-67).

BCG Sensor. The BCG signals were acquired using a custom mattress-mounted system developed by BOBO Technology Co., Ltd. (Fig. 1(c)). The system employs a polyvinylidene fluoride (PVDF) piezoelectric film sensor (dimensions: 80 cm $\times$ 5 cm; sampling rate: 125 Hz) with a piezoelectric coefficient of 19 $\pm$ 4 pC/N and capacitance of 10 nF. The PVDF sensor's high sensitivity and mechanical durability ensure reliable conversion of body movements into electrical signals, even during prolonged nocturnal monitoring.

The piezoelectric effect of PVDF film is attributed to the polar response characteristics of its lattice structure[48]. When the left ventricle of the human heart ejects blood during the systolic phase, the resulting change in blood momentum creates an instantaneous mechanical disturbance along the head-foot axis. During diastole, the filling of the ventricles and the elastic retraction of the vessels generate reverse mechanical fluctuations[49]. These mechanical signals travel through the bones and soft tissues to the body surface, causing mechanical deformation of the PVDF film. According to the principle of the piezoelectric effect[50], the surface charge (Q) of the sensor has a linear relationship with the applied force, which can be mathematically expressed in Eq. (1).

$$q=\sum(d_{ij}\delta_i)=\sum d_{ij}\frac{F_i}{A_i}. \tag{1}$$

where q is the number of charges induced in different directions, d_{ij} is the piezoelectric coefficient matrix, and δ_i is the unit force applied to the area A_i.

ECG Sensor. In this study, we utilized a three-lead Holter device (Fig. 1(d), model: Beneware CT-08S) for simultaneous ECG and BCG acquisition. The device has several important technical specifications: it features a sampling rate of 200 Hz, a resolution of 12 bits, and a dynamic input range of $\pm$10 mV. Additionally, its frequency response range is $\pm$3 dB@0.05-60 Hz, and it operates on a power supply voltage of 1.5 V $\pm$ 10%. The device weighs approximately 80 g and supports data transmission via an external SD card reader or USB interface.

After the ECG is collected, the data is uploaded onto the Beneware intelligent analysis platform for further processing. This platform features a comprehensive ECG analysis function that can automatically extract key ECG parameters, such as HR, and intelligently identify abnormal rhythms, including atrial premature beats and ventricular arrhythmia. It can also accurately detect morphological changes, such as shifts in the ST-T segment.

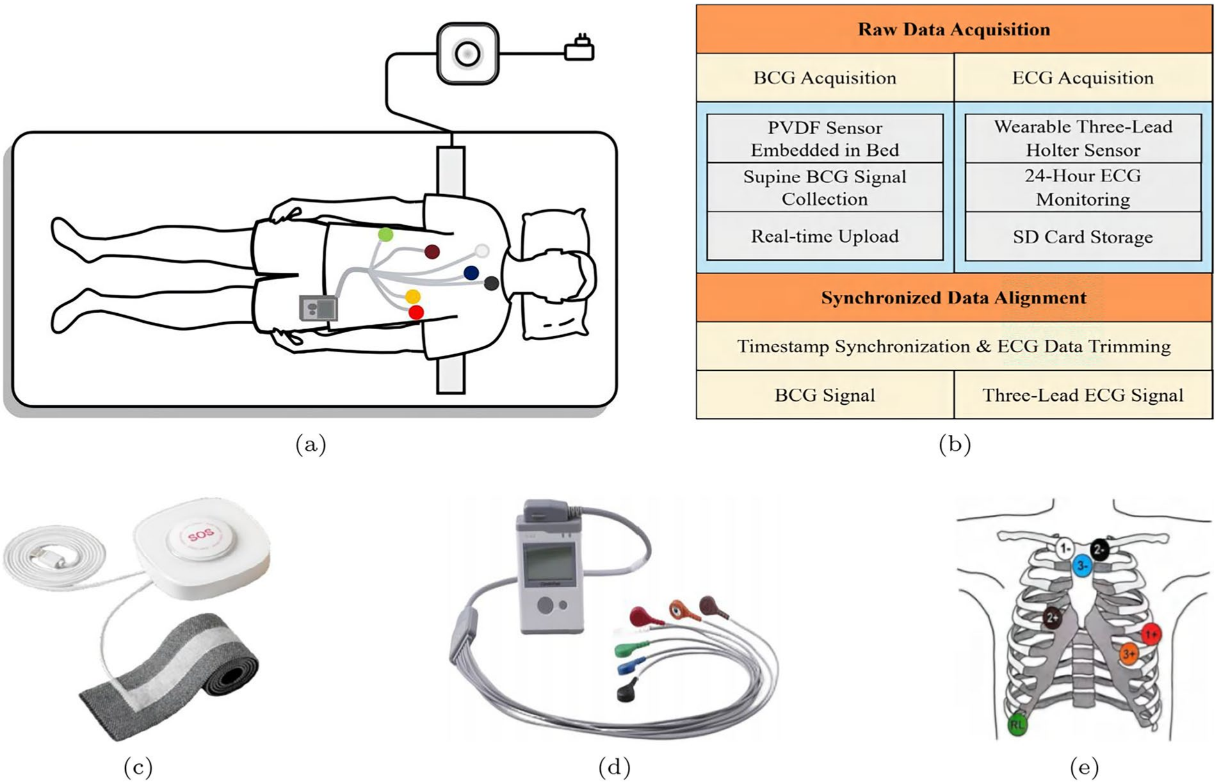


Fig. 1 The data collection scheme implemented in this study: (**a**) placement of the sensor, (**b**) BCG/ECG data acquisition and alignment workflow, (**c**) device for acquiring BCG signals, (**d**) synchronous Holter monitor, and (**e**) electrode placement of the three-lead Holter.

Electrode Definition	Identification	Color	Lead	Reference Position
Positive Pole of Channel 1	1+	Red		Intersection of the left anterior axillary line and the fifth intercostal space
Negative Pole of Channel 1	1−	White	Simulated V5	Intersection of the right clavicle and sternum
Positive Pole of Channel 2	2+	Brown		Intersection of the right sternal border and the fourth intercostal space
Negative Pole of Channel 2	2−	Black	Simulated V1	Intersection of the left clavicle and sternum
Positive Pole of Channel 3	3+	Orange		Intersection of the left midclavicular line and the sixth rib
Negative Pole of Channel 3	3−	Blue	Simulated V3	Sternum handle
Common Electrode	RL	Green	—	Right chest wall and lower edge of the rib

Table 4. Electrode vonfiguration for three-lead holter.

To ensure the reliability of the diagnostic results, all analysis outcomes are reviewed and approved by specialists in the ECG room for the clinical accuracy and diagnostic value of the data. The overall data collection and processing flow is illustrated in Fig. 1(b).

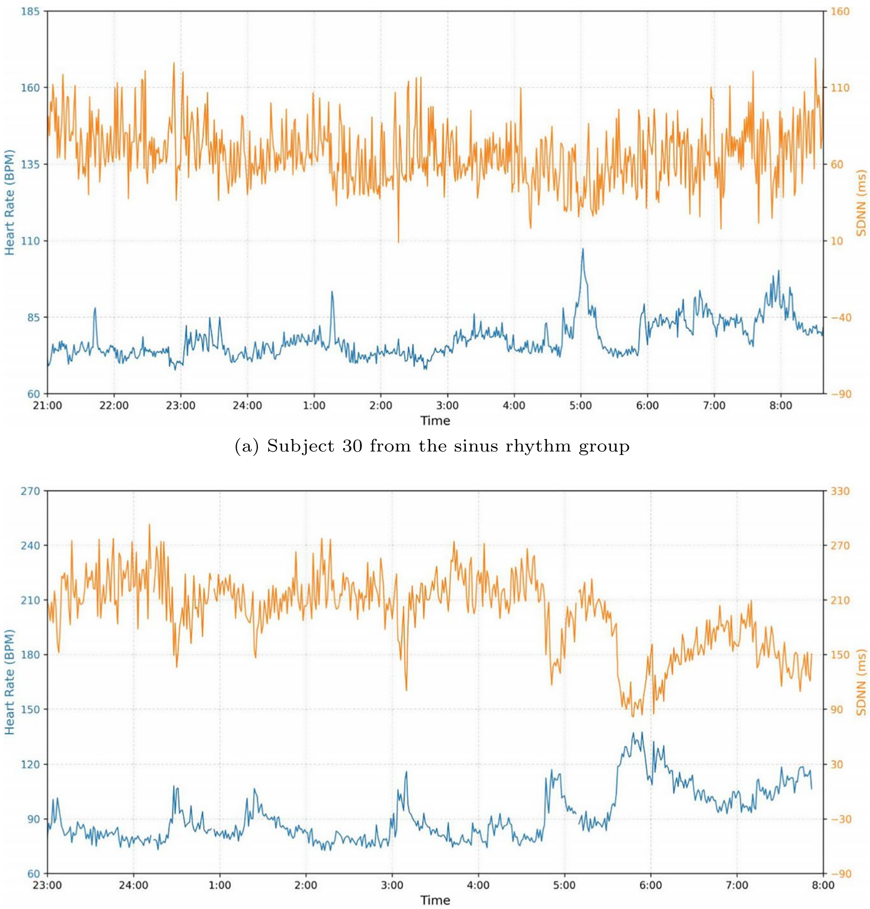
Procedures. The data collection took place at Run Run Shaw Hospital, which is affiliated with Zhejiang University School of Medicine. All participants were either admitted for a health check or underwent 24-hour ECG monitoring due to health concerns. The Holter sensor was worn according to the instrument's specifications, using 7 electrodes and 3 lead wires for ECG collection. The definitions of each lead wire and its reference connections are provided in Fig. 1(e) and Table 4. The ECG collection for each subject begins between 4 and 6 p.m. and continues uninterrupted until the same time the following day, ensuring approximately 24 hours of ECG monitoring.

The BCG sensor belt is securely positioned horizontally on the surface of the mattress, with its long axis aligned perpendicular to the head-to-foot direction of the person lying down (Fig. 1(a)). This orientation ensures that the heart position of the supine subject corresponds vertically to the sensor belt. After the nurse has completed the equipment setup by following standard procedures, a bed sheet is placed over the belt to ensure physical stabilization to prevent the sensor from being displaced due to the subject's movements.

Data collection occurs from the time of bedtime preparation until the subject wakes up the following day. A designated person manually starts and stops the device, carefully recording the precise timing. The sensor data

	All (n=46 Male=56.52%)		Sinus (n=29 Male= 35.29%)		Arrhythmic (n=17 Male= 68.96%)	
	Mean & STD	**Range**	**Mean & STD**	**Range**	**Mean (STD)**	**Range**
Age(years)	75.75 ± 15.09	[27,93]	72.74 ± 17.44	[27,92]	80.15 ± 9.84	[59,93]
Heights(cm)	164.44 ± 8.82	[150.0,179.0]	166.74 ± 8.1	[154.0,179.0]	161.08 ± 9.06	[150.0,175.0]
Weight (Kg)	56.34 ± 11.72	[40.0,82.0]	57.37 ± 9.42	[40.0,70.0]	54.85 ± 14.74	[40.0,82.0]
BMI	20.67 ± 3.12	[16.02,30.12]	20.49 ± 2.02	[16.65,23.01]	20.92 ± 4.36	[16.02,30.12]
HR (bpm)	69.95 ± 11.28	[49.68,94.76]	67.22 ± 10.06	[51.83,94.76]	74.61 ± 12.02	[49.68,90.92]
SDNN	134.72 ± 61.85	[49.83,317.49]	107.21 ± 42.45	[49.83,225.61]	181.66 ± 62.39	[61.77,317.49]
RMSSD	131.38 ± 101.78	[19.3,435.82]	72.82 ± 48.66	[19.3,227.54]	231.26 ± 90.43	[70.68,435.82]
pNN50	34.66 ± 32.34	[0.36,89.39]	13.53 ± 15.77	[0.36,66.26]	70.69 ± 17.83	[21.79,89.39]

Table 5. Baseline information for the subjects.



(a) Subject 30 from the sinus rhythm group

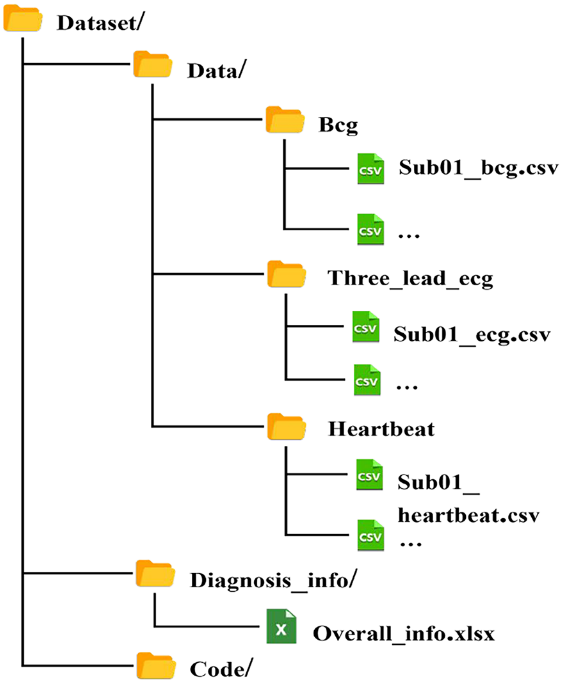
(b) Subject 6 with diagnosed atrial fibrillation

Fig. 2 Long-term trends of cardiac activities for two subjects with different health conditions: (**a**) Subject 30 from the sinus rhythm group (**b**) Subject 6 with diagnosed atrial fibrillation.

is uploaded to the cloud-end in real-time via encrypted transmission, while the corresponding ECG signals are extracted based on the recorded time. Despite synchronizing multiple device clocks using the Network Time Protocol (NTP) protocol prior to the experiment, manual calibration was ultimately performed to ensure the accurate alignment between signal timing and to address any drift between the devices.

Table 5 summarizes the statistical information for the subjects, including their demographic data and the cardiac rhythmic indices obtained from the reference ECG signals. While the two groups of individuals, including those with sinus rhythm and arrhythmia patients, share similar demographic distributions, their rhythmic indices exhibit distinguishable patterns.

To ensure the accuracy of data labeling, experienced experts conducted a thorough analysis of the synchronous ECG data from all subjects using the Beneware intelligent analysis platform. The ECG annotation process ensured rigor through multiple measures. First, two senior specialists with over 10 years of experience adhered strictly to the AHA/ACCF/HRS 2009 Recommendations. They followed precise definitions for core waveforms



(a)

time	value
23:00:02	1442
23:00:03	2269
……	……

(b)

time	c1	c2	c3
23:00:02	2106	1909	16
23:00:03	2224	1865	16
……	……	……	……

(c)

Beat Symbol	RR Interval	time
N	802	22:59:58.446
S	472	23:00:18.602
V	462	23:00:01.157
X	-	23:11:02.250

(d)

Idx	Sex	Age	Height	Weight	Conclusion	ST Segment Information	Atrial Fibrillation
1	M	91	170cm	50kg	Recording period: 0-0-1 16:46:12 to 0-0-2 08:06:24. 1. Persistent atrial fibrillation, mostly accompanied by normal ventricular rate (38~198 bpm, average 70 bpm). Further evaluation and treatment in the atrial fibrillation clinic are recommended. ……	Maximum Depression(mV): -0.432 Event Start Time:0-0-2 00:04:24 Event Duration(minutes):10.8	/
…	……	……	……	……	……	……	……
34	F	59	/	/	Recording Time: From 0-0-1 09:17:32 to 0-0-2 08:22:50 Persistent Atrial Fibrillation throughout, accompanied by a normal ventricular rate (78-135 beats per minute, average 84 beats per minute). ……	Maximum Depression(mV):-0.119 Event Start Time: 0-0-2 06:08:30 Event Duration(minutes):1.11 ST Segment Slope:DownSloping ……	Event Start Time: 0-0-2 08:17:32 Event End Time: 0-0-2 08:22:50 Event Type:Atrial Fibrillation/Atrial Flutter
…	……	……	……	……	……	……	……

(e)

Fig. 3 (**a**) Overall Structure of the Dataset. (**b**) Structure of BCG Data File (**c**) Structure of ECG Data File (**d**) Structure of Heartbeat File and (**e**) Structure of Overall_info File.

and key parameters, with clear diagnostic criteria to determine conditions. A multidimensional evaluation prevented bias by integrating ECG morphology (e.g., arrhythmic rhythms, ischemic ST changes) with clinical context, including subjects' medical histories, medications, and symptom timelines, for comprehensive interpretations. When discrepancies arose between annotators, a senior physician with over 20 years of experience made the final judgment. These measures ensured annotation accuracy, reinforcing the dataset's reliability.

To illustrate the variations of subjects' cardiac activities during overnight monitoring, we selected two representative subjects: one control subject with no known history of arrhythmia (subject ID 30) and one patient with a confirmed case of atrial fibrillation (subject ID 6). Fig. 2 shows the dynamic trends of heart rate and SDNN (standard deviation of heart rate variability) using a 1-minute time granularity. It reflects the fluctuation patterns of the key indices over time.

Data Record

The dataset is available on FigShare (https://doi.org/10.6084/m9.figshare.28643153)[51] and consists of several folders: Data, Diagnosis_info, and Code. The structure of this dataset is illustrated in Fig. 3(a).

The Data folder contains information from 46 patients, which is further divided into three subfolders: Bcg, Three_lead_ecg, and Heartbeat. To protect patient privacy, each patient is identified by a unique number

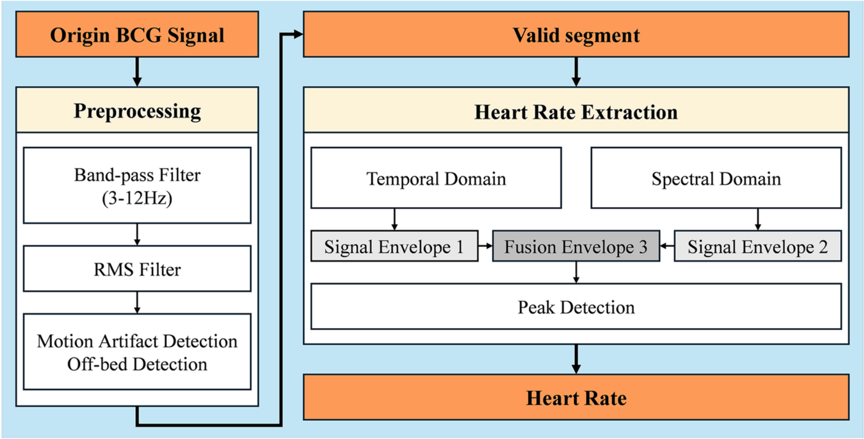


Fig. 4 The HR extraction process based on BCG signals.

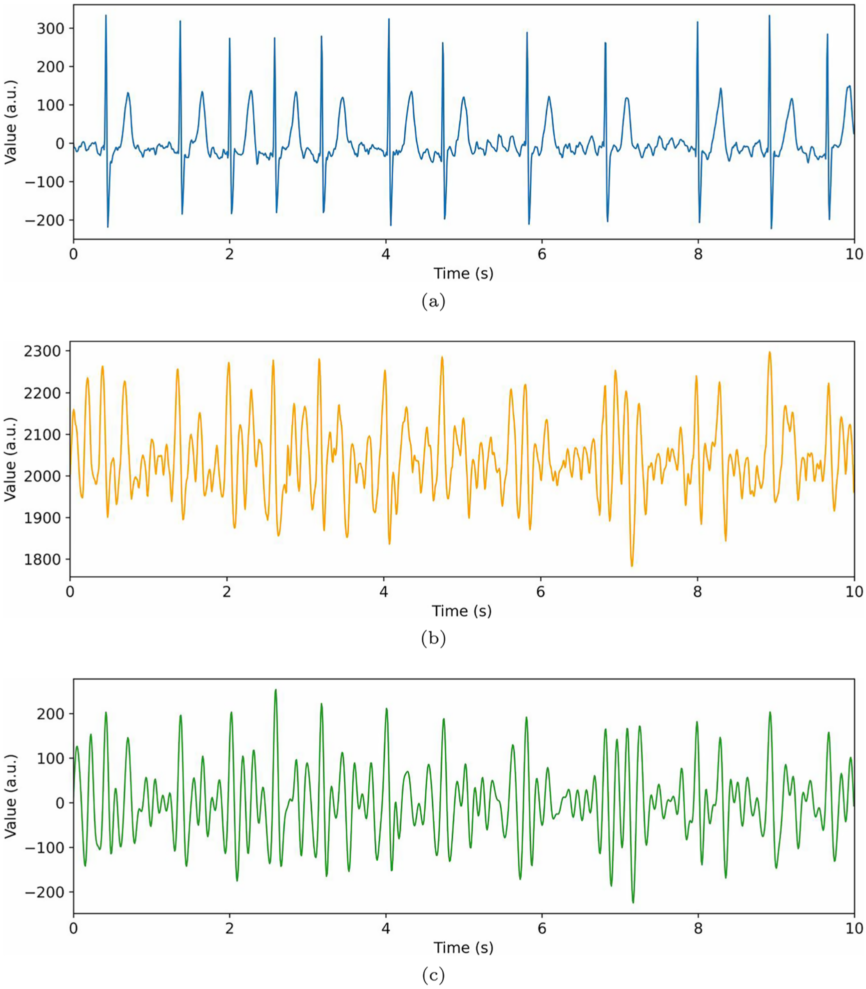


Fig. 5 (**a**) Synchronized ECG signal with atrial fibrillation (**b**) Raw BCG signal and (**c**) Bandpass-filtered (3-12Hz) BCG signal.

(SubID), and specific year and month information for all timestamps has been removed; only hours, minutes, and seconds are retained. For data records that may span different time contexts, it is recommended to make reasonable inferences based on the surrounding information during analysis. A detailed description of each document is provided as follows:

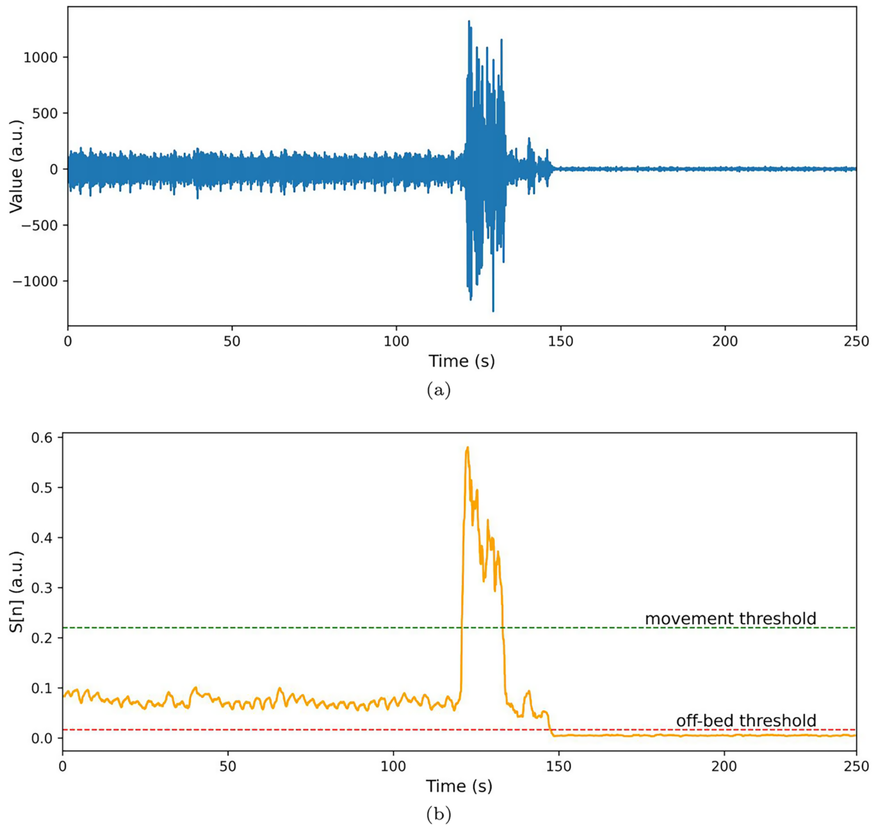


Fig. 6 (**a**) BCG signal. (**b**) Corresponding RMS values for the segments shown in (**a**).

The files in the Bcg subfolder are named using the format SubID_bcg.csv and contain two columns of data (Fig. 3(b)): column (a) is a timestamp, and column (b) is a BCG signal recorded at a sampling rate of 125 Hz. The file in the Three_lead_ecg subfolder is named SubID_ecg.csv and includes four columns of data (Fig. 3(c)). These correspond to the timestamp and the standard limb lead electrocardiogram (ECG) signals for leads I, II, and III, recorded at a sampling rate of 200 Hz.

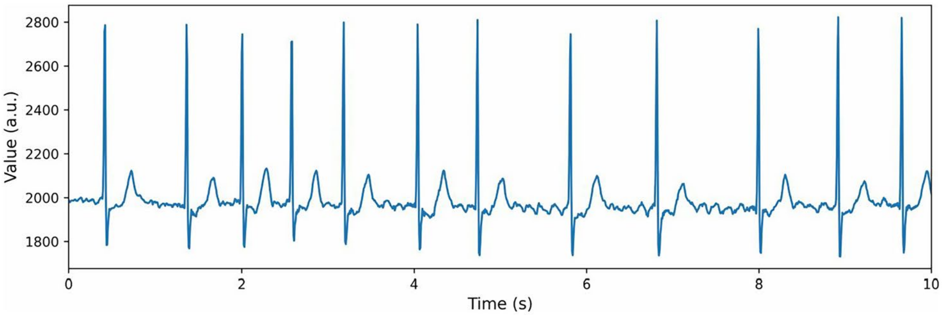
The files in the Heartbeat subfolder are named SubID_heartbeat.csv and contain three columns of data (Fig. 3(d)). Column (a) indicates the type of arrhythmia event, with classifications as follows: V for a ventricular premature beat, S for an atrial premature beat, N for a normal heartbeat, and X for pseudotemplate (a non-physiological signal introduced by equipment failure, power interference, motion artifact, or poor electrode contact). Column (b) represents the RR interval, and column (c) is the timestamp.

Additionally, we have a Diagnosis_info folder containing Overall_info.xlsx files that detail basic information and diagnostic findings for all subjects (Fig. 3(e)). This includes gender, age, height, weight, diagnosis, and diagnostics. Events such as atrial fibrillation, premature beats, and abnormal ST segments occurred during the recording of BCG data.

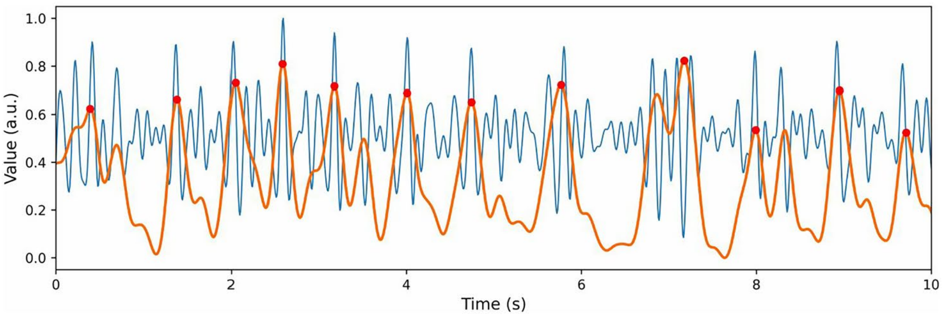
Finally, the Code folder contains scripts that validate the quality of the dataset.

Technical Validation

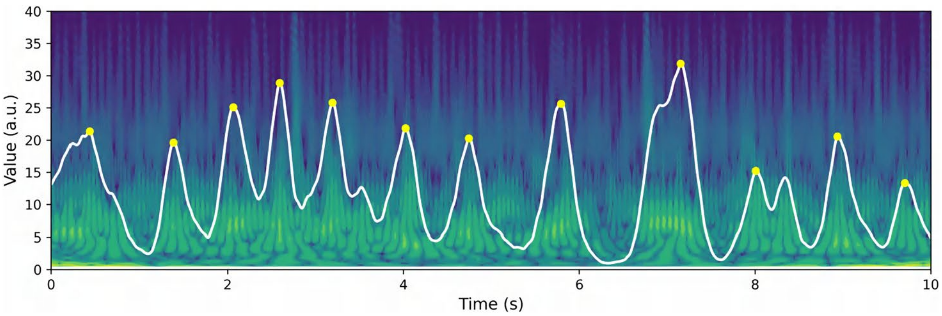
When the BCG sensor collects the impulse signal, it not only registers the mechanical vibrations generated by the heartbeat but is also affected by various sources of interference. These include low-frequency harmonic components produced by breathing movements, high-frequency noise introduced by the sensor circuit, and time-domain interference due to changes in the subject's position. In a typical nighttime bedridden environment, movements can cause two main types of disturbances: transient amplitude distortion from body movement, which can obscure the effective cardio-impulse waveform, and incidents of getting out of bed or poor sensor contact that can lead to a sharp drop in signal amplitude. These complex interferences significantly diminish the signal-to-noise ratio of the original BCG signal. Therefore, it is essential to implement noise suppression techniques and extract relevant signal fragments through various preprocessing methods. Fig. 4 illustrates the specific processing steps of our HR extraction approach[52] proposes a bandpass filter's range of 3-10 Hz for the quick identification of heartbeats based on analysis of BCG frequency harmonic components. Considering that the cohort of this work involves patients with CVD, we expanded the band and applied a 3-12 Hz bandpass filter to the original BCG signal after empirical studies. This filtering approach aims to retain the active components closely related to the cardiac cycle while removing respiratory harmonic components and noise generated by the sensor, as shown in Fig. 5.



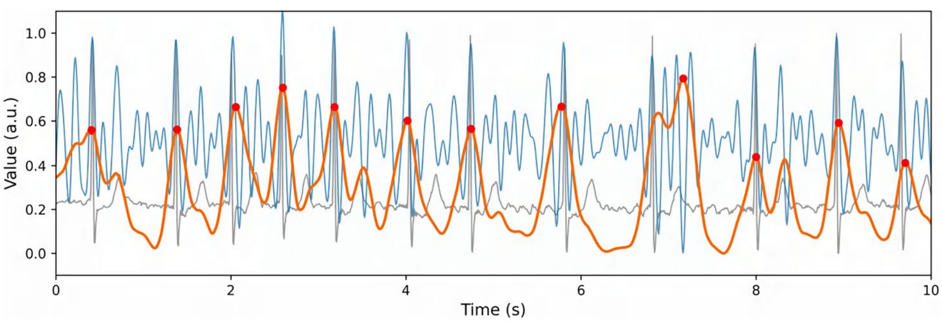
(a)



(b)



(c)



(d)

Fig. 7 (**a**) Synchronized ECG signal with atrial fibrillation (**b**) Time-domain envelope & Peak Point (**c**) Spectral-domain Envelope & Peak Point and (**d**) Time-Frequency domain fusion envelope, Peak Point and Synchronized ECG signals. Note that the phenomenon of heartbeat recognition point offset occurs in the fourth-to-last heartbeat. Future studies could explore the integration of semantic context to enhance detection accuracy.

Signal Quality Assessment and Thresholding. To identify invalid BCG segments (e.g., motion artifacts, off-bed events, or poor sensor contact), we applied a root mean square (RMS) filter, building on methods from prior studies[53]. The RMS signal S[n] was computed as Eq. (2).

Sub_Id	Valid Segment Proportion	Sub_Id	Valid Segment Proportion	Sub_Id	Valid Segment Proportion
1	83.06%	17	57.80%	33	57.04%
2	86.87%	18	89.19%	34	78.19%
3	84.71%	19	56.07%	35	52.63%
4	94.20%	20	74.75%	36	65.78%
5	80.50%	21	56.54%	37	63.91%
6	69.29%	22	54.70%	38	76.55%
7	84.43%	23	79.83%	39	66.45%
8	96.61%	24	55.23%	40	60.59%
9	70.57%	25	74.35%	41	64.43%
10	84.70%	26	77.31%	42	72.71%
11	89.05%	27	54.44%	43	61.99%
12	83.75%	28	84.41%	44	58.76%
13	82.89%	29	57.51%	45	74.01%
14	76.88%	30	76.13%	46	60.01%
15	85.38%	31	36.56%		
16	84.75%	32	75.95%		

Table 6. Valid segment proportions for individual participants. Average Availability Percentage: 70.95(%) Average Effective Duration: 443.16(minutes).

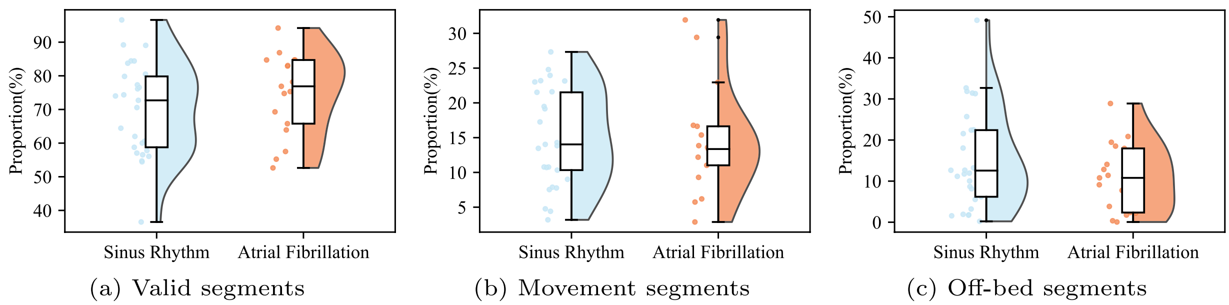


Fig. 8 Distributions of the segment states by sinus rhythm subjects and by atrial fibrillation patients: (**a**) Valid segments (**b**) Movement segments (**c**) Off-bed segments.

$$S[n] = \sqrt{\frac{1}{N}\sum_{k=0}^{N-1}(X[n+k])^2} \tag{2}$$

where X[n] is the raw BCG signal, S[n] is the smoothed RMS output, and N=200 (1.6 seconds at 125 Hz sampling rate). As shown in Fig. 6, valid BCG segments exhibit stable S[n] fluctuations, whereas motion artifacts ($S[n] > \tau_2$) and off-bed events ($S[n] < \tau_1$) deviate markedly. We classified segments using Eq. (3):

$$State(i) = \begin{cases} Movement, & if\ \ S[n] > \tau_2 \\ Valid, & if\ \ \tau_1 \leq S[n] \leq \tau_2 \\ Off\text{-}bed, & if\ \ S[n] < \tau_1 \end{cases} \tag{3}$$

To balance interindividual physiological variability with experimental practicality, thresholds τ_1 and τ_2 were standardized as follows:

- Normalize S[n] values to [0, 1] for each subject.
- Compute the median S[n] from 20 stable 200-second segments per subject.
- Set $\tau_1 = 0.25 \times median$ and $\tau_2 = 2 \times median.$

This approach accommodates baseline physiological differences while ensuring consistent segmentation across participants.

Following signal segmentation, we excluded all epochs labeled as body movement, out-of-bed activity, and their 5-second buffer periods, retaining only continuous BCG segments ≥5 minutes for HR analysis. While prior BCG-based HR extraction methods[24] achieved promising results, they often rely on assumptions of stable

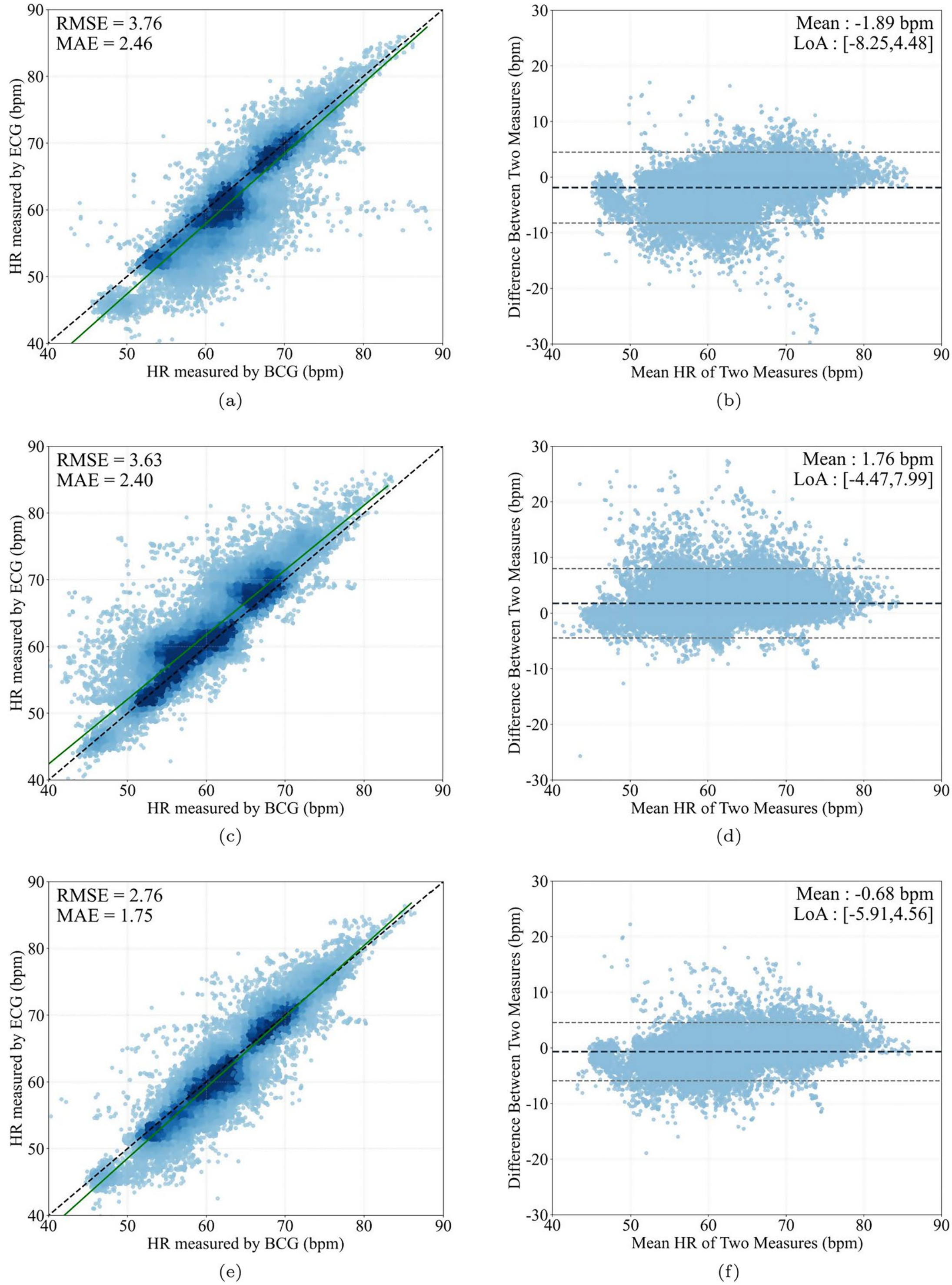


Fig. 9 (**a**),(**c**),(**e**) Scatter density plots on Time-domain, Frequency-domain, Time-Frequency Fusion; (**b**),(**d**),(**f**) Bland-Altman plots on Time-domain, Frequency-domain, Time-Frequency Fusion.

IBIs in healthy individuals. To address this limitation, we eschewed complex physiology-dependent algorithms in favor of a time-frequency fusion approach:

Feature Extraction. BCG signals were mapped to HR features using parallel morphological (time-domain energy envelope)[21,23] and spectral (frequency-domain energy envelope)[31] analyses. In the time domain, the Hilbert transform was employed for envelope extraction, while the frequency domain utilized the wavelet transform. The resulting envelopes were then linearly combined to generate a composite signal.

Peak Detection. The automatic multi-scale peak detection (AMPD) algorithm[54] identified cardiac-related peaks in the composite envelope (Fig. 7).

Validation. HR was calculated as the number of detected peaks per 60-second segment and compared against ECG-derived rates (from R-R intervals).

This dual-channel strategy enhances robustness in low-noise conditions by synergizing time-domain J-peak prominence and frequency-domain signal-to-noise ratio (SNR), allowing reliable extraction of heart rhythmic indices without restrictive physiological assumptions.

Table 6 summarizes the dataset's signal quality, with valid BCG segments accounting for >70% of total recording time (mean = 443.16 minutes) across all participants. This demonstrates the dataset's utility for deriving reliable cardiovascular metrics. While the current fixed segmentation thresholds (τ_1, τ_2) enable robust group-level analysis, interindividual variability in physiological signals suggests opportunities for optimization. Implementing adaptive threshold algorithms, which dynamically adjust τ_1 and τ_2 based on real-time trends in resting-state signal quality $S[n]$, could enhance both the temporal resolution of segmentation and the yield of valid segments for individual participants.

Figure 8 illustrates the distributions of BCG segment proportions by participants across different states. Although subjects with sinus rhythm and patients with atrial fibrillation have similar means for all three states, their variances in distribution are quite different. Generally, the distributions for participants with sinus rhythm show a greater degree of dispersion. We notice that, while one subject (with Sub_ID 31)with sinus rhythm deviates significantly in off-bed time and leads to the corresponding valid state dropping below 40%, there are more outliers among patients with atrial fibrillation in the movement state.

Figure 9 compares BCG-derived HR estimates against ECG (the clinical gold standard) using three signal-processing strategies: time-series energy envelope (a), frequency-domain energy envelope (c), and time-frequency composite envelope (e). Each panel pairs a density scatter plot (left) with a Bland-Altman agreement plot (right). To mitigate overplotting, data points in the scatter plots are jittered with minor random offsets. The composite envelope method (e) demonstrates superior agreement, with points tightly clustered around the identity line (y = x) in the scatter plot and minimal bias in the Bland-Altman analysis.

The scatter density plots (Fig. 9(a),(c),(e)) illustrate the agreement between BCG- and ECG-derived HR. Heatmap color gradients denote data density: dark colors (e.g., black) indicate high-density regions, while light colors (blue) reflect sparser distributions. The dashed black line represents the ideal identity line (y=x), and the green curve shows the empirical regression fit. Among the three methods tested, the composite time-frequency envelope (Fig. 9(e)) demonstrates the tightest clustering around the identity line, with 95.6% of points within the 95% limits of agreement. Its regression equation ($y = 1.06x \pm 4.61$) aligns closely with the ideal trend, supported by a strong Pearson correlation ($r = 0.92$, $p < 0.001$) and low error metrics (RMSE = 2.76 bpm, MAE = 1.75 bpm).

These results meet the Association for the Advancement of Medical Instrumentation (AAMI) clinical standards for HR monitors, which permit errors of ≤10% or ±5 bpm[55]. This preliminary estimation validates BCG's reliability for HR estimation using simple conventional algorithms, providing a solid foundation for enhancing the assessment of cardiac rhythm indices (e.g., HRV, IBI) through more advanced machine learning methods.

The Bland-Altman plots (Fig. 9(b),(d),(f)) assess agreement between BCG and ECG derived HR measurements. The horizontal axis represents the arithmetic mean of paired BCG/ECG HRs, while the vertical axis shows their differences. The dashed black line denotes the mean bias (average difference), and the upper/lower dashed lines mark the 95% limits of agreement (LoA). For time-series envelope-derived HRs (Fig. 9(d)), 95.6% of data points fall within the LoA, compared to 95.3% for frequency-domain envelope methods (Fig. 9(e)). The combined time-frequency fusion approach (Fig. 9(f)) achieves narrower difference distributions and reduced mean bias compared to single-feature methods, demonstrating the enhanced robustness of the fusion strategy.

Code availability

The signal processing pipeline, including filtering, valid segment detection, envelope extraction, and keypoint identification, was implemented in Python 3.12. All algorithms, visualization scripts, and reproducible analysis workflows are publicly available in the GitHub repository https://github.com/hustselab511/DatasetTechnicalValidation. While Python is recommended for compatibility, the dataset and code can be adapted to other environments (e.g., MATLAB).

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Competing interests

The authors declare no competing interests.

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